

EXPONENTIALS AND LOGARITHMS

- 1 Express in the form $p \log_{10} a + q \log_{10} b$
 - a $\log_{10} ab$
 - b $\log_{10} ab^7$
 - c $\log_{10} \frac{a^3}{b}$
 - d $\log_{10} a\sqrt{b}$
 - e $\log_{10} (ab)^2$
 - f $\log_{10} \frac{1}{ab}$
 - g $\log_{10} \sqrt{a^3b^5}$
 - h $3 \log_{10} \frac{a^2}{\sqrt[3]{b}}$
- 2 Given that $y = \log_q 8$, express each of the following in terms of y .
 - a $\log_q 64$
 - b $\log_q 2$
 - c $\log_q \frac{16}{q}$
 - d $\log_q 4q^3$
- 3 Given that $a = \lg 2$ and $b = \lg 3$, express each of the following in terms of a and b .
 - a $\lg 18$
 - b $\lg 96$
 - c $\lg \frac{9}{16}$
 - d $\lg 6 - \lg 8$
 - e $\lg \sqrt{6}$
 - f $\frac{3}{2} \lg 16 + \frac{1}{2} \lg 81$
 - g $4 \lg 3 - 3 \lg 6$
 - h $\lg 60 + \lg 20 - 2$
- 4 Without using a calculator, evaluate
 - a $\frac{1}{3} \log_5 1000 - \frac{1}{2} \log_5 4$
 - b $2 \log_{12} 4 + \frac{1}{2} \log_{12} 81$
 - c $\log_4 12 + \log_4 \frac{2}{3}$
 - d $\frac{\log_7 81}{\log_7 3}$
 - e $3 \log_{27} 12 - 2 \log_{27} 72$
 - f $\frac{\log_{11} 25}{\log_{11} \frac{1}{5}}$
- 5 Solve each equation, giving your answers correct to 3 significant figures.
 - a $\log_3 x = 1.8$
 - b $\log_5 x = -0.3$
 - c $\log_8 (x - 3) = 2.1$
 - d $\log_4 (\frac{1}{2}x + 1) = 3.2$
 - e $15 - \log_2 3y = 9.7$
 - f $\log_6 (1 - 5t) + 4.2 = 3.6$
- 6 Express in the form $\log_2 [f(x)]$
 - a $5 \log_2 x$
 - b $\log_2 x + \log_2 (x + 4)$
 - c $2 \log_2 x + \frac{1}{5} \log_2 x^5$
 - d $3 \log_2 (x - 2) - 4 \log_2 x$
 - e $\log_2 (x^2 - 1) - \log_2 (x + 1)$
 - f $\log_2 x - \frac{1}{2} \log_2 x^4 + \frac{1}{3} \log_2 x^2$
- 7 Solve each of the following equations.
 - a $\log_3 x + \log_3 5 = \log_3 (2x + 3)$
 - b $\log_9 x + \log_9 10 = \frac{3}{2}$
 - c $\log_4 x - \log_4 (x - 1) = \log_4 3 + \frac{1}{2}$
 - d $\log_5 5x - \log_5 (x + 2) = \log_5 (x + 6) - \log_5 x$
 - e $2 \log_6 x = \log_6 (2x - 5) + \log_6 5$
 - f $\log_7 4x = \log_7 \frac{1}{x-6} + 1$
- 8 Solve each pair of simultaneous equations.
 - a $\log_x y = 2$
 $xy = 27$
 - b $\log_5 x - 2 \log_5 y = \log_5 2$
 $x + y^2 = 12$
 - c $\log_2 x = 3 - 2 \log_2 y$
 $\log_y 32 = -\frac{5}{2}$
 - d $\log_y x = \frac{3}{2}$
 $x^{\frac{1}{3}} + 3y^{\frac{1}{2}} = 20$
 - e $\log_a x + \log_a 3 = \frac{1}{2} \log_a y$
 $3x + y = 20$
 - f $\log_{10} y + 2 \log_{10} x = 3$
 $\log_2 y - \log_2 x = 3$